

All Book Questions

Chapter 1

Introduction to Visual C#

Question 1

A(n) _____ is the thin dotted line that encloses an object in the Designer.

Choices:

- a. control binder
- b. bounding box
- c. object container
- d. form outline

Correct Answer:

b. bounding box

Explanation:

The bounding box is the dotted outline that appears around a control or form in the Designer, showing its boundaries.

Question 2

The small squares that appear on the edges of a form's bounding box are called _____.

Choices:

- a. sizing hooks
- b. form edges
- c. bounding tags
- d. sizing handles

Correct Answer:

d. sizing handles

Explanation:

Sizing handles allow the developer to resize a form or control by clicking and dragging.

Question 3

_____ is the name of the blank form that Visual Studio initially creates in a new project.

Choices:

- a. Form1
- b. Main
- c. New1
- d. Blank

Correct Answer:

a. Form1

Explanation:

Visual Studio names the first form in a new project “Form1” by default.

Question 4

The _____ property holds the text that is displayed on the face of a button.

Choices:

- a. Name
- b. Text
- c. Tag
- d. Face

Correct Answer:

b. Text

Explanation:

The Text property determines the text shown on controls such as Buttons and Labels.

Question 5

A file that contains program code is called a(n) _____.

Choices:

- a. destination code file
- b. executable file
- c. machine language file
- d. source code file

Correct Answer:

d. source code file

Explanation:

Source code files contain human-readable code written by the programmer before compilation.

Question 6

A namespace is a container that holds _____.

Choices:

- a. methods
- b. names
- c. spaces
- d. classes

Correct Answer:

d. classes

Explanation:

Namespaces organize and group related classes to avoid naming conflicts.

Question 7

A(n) _____ is a method that executes when a specific event takes place while an application is running.

Choices:

- a. action process
- b. event handler
- c. runtime procedure
- d. event method

Correct Answer:

b. event handler

Explanation:

An event handler runs automatically in response to an event such as a button click.

Question 8

The statement `MessageBox.Show("Hello World");` is an example of a(n) _____.

Choices:

- a. method call
- b. namespace
- c. Click event
- d. event handler

Correct Answer:

a. method call

Explanation:

This statement calls the Show method of the MessageBox class.

Question 9

In programming, the term string means _____.

Choices:

- a. many lines of code
- b. parallel memory locations
- c. string of characters
- d. virtually anything

Correct Answer:

c. string of characters

Explanation:

A string is a sequence of characters such as letters, numbers, or symbols.

Question 10

A(n) _____ marks the end of a programming statement in C#.

Choices:

- a. semicolon
- b. period
- c. hyphen
- d. underscore

Correct Answer:

a. semicolon

Explanation:

Most C# statements must end with a semicolon (;).

Question 11

A piece of data that is written directly into a program's code is called a(n) _____.

Choices:

- a. identifier
- b. specifier
- c. keyword
- d. literal

Correct Answer:

d. literal

Explanation:

A literal is a fixed value written directly in the code, such as 5 or "Hello".

Question 12

The time during which you build the GUI and write the application's code is referred to as _____.

Choices:

- a. run time
- b. design time
- c. code time
- d. planning

Correct Answer:

b. design time

Explanation:

Design time is when the program is being created, not executed.

Question 13

The time during which an application is executing is referred to as _____.

Choices:

- a. go time
- b. design time
- c. execution
- d. run time

Correct Answer:

d. run time

Explanation:

Run time refers to when the program is actively running.

Question 14

When you want to display text on a form, you use a _____ control.

Choices:

- a. Button
- b. PictureBox
- c. Label
- d. TextBox

Correct Answer:

c. Label

Explanation:

A Label control is designed to display non-editable text.

Question 15

The _____ property allows you to set the font, font style, and size of a control's text.

Choices:

- a. Style
- b. AutoSize
- c. Text
- d. Font

Correct Answer:

d. Font

Explanation:

The Font property controls how text appears in a control.

Question 16

A _____ property can be set to one of two possible values: True or False.

Choices:

- a. Boolean
- b. Logical
- c. Binary
- d. Dual

Correct Answer:

a. Boolean

Explanation:

Boolean properties store only True or False values.

Question 17

Label controls have a(n) _____ property that controls the way they can be resized.

Choices:

- a. Stretch
- b. AutoSize
- c. Dimension
- d. Fixed

Correct Answer:

b. AutoSize

Explanation:

When AutoSize is True, the Label resizes automatically to fit its text.

Question 18

The _____ property can be used to change the text alignment in a Label.

Choices:

- a. TextPosition
- b. AutoAlign
- c. TextCenter
- d. TextAlign

Correct Answer:

d. TextAlign

Explanation:

TextAlign controls where the text appears inside the Label.

Question 19

In code, you use a(n) _____ to store a value in a control's property.

Choices:

- a. Click event

- b. method call
- c. assignment statement
- d. Boolean value

Correct Answer:

c. assignment statement

Explanation:

Assignment statements use the = operator to store values.

Question 20

The equal sign (=) is known as the _____.

Choices:

- a. equality symbol
- b. assignment operator
- c. equality operator
- d. property position

Correct Answer:

b. assignment operator

Explanation:

The assignment operator assigns the value on the right to the variable or property on the left.

True / False

Statement 1

Changing an object's Text property also changes the object's name.

Correct Answer: False

Explanation:

The Name and Text properties are independent. Changing Text does not affect Name.

Statement 2

When a form is created, its Text property is initially set to the same value as the form's name.

Correct Answer: True

Explanation:

By default, the form's title matches its name.

Statement 3

The form's title is displayed in the bar along the top of a form.

Correct Answer: True

Explanation:

This area is called the title bar.

Statement 4

C# source code files always end with the .cs extension.

Correct Answer: True

Explanation:

.cs is the standard extension for C# source files.

Statement 5

You add your own code to the Program.cs file as you develop an application.

Correct Answer: False

Explanation:

Most user code is written in form or class files, not Program.cs.

Statement 6

C# code is organized as methods, which are contained inside classes, which are contained inside namespaces.

Correct Answer: True

Explanation:

This is the standard structural hierarchy in C#.

Statement 7

In C# code, each opening brace must have a corresponding closing brace.

Correct Answer: True

Explanation:

Braces define code blocks and must always be paired.

Statement 8

When you double-click a control in the Designer, Visual Studio creates an event handler and supporting code.

Correct Answer: True

Explanation:

Visual Studio automatically generates required code for the event to function.

Statement 9

A Label control's Text property is initially set to the same value as the Label control's name.

Correct Answer: True

Explanation:

Labels are created with default text matching their name.

Statement 10

When a Label control's AutoSize property is set to True, you cannot manually resize it.

Correct Answer: True

Explanation:

AutoSize overrides manual resizing.

Chapter 2

Processing Data (Variables, Data Types, and Calculations)

Question 1

When the user types into a TextBox control, the text is stored in the control's _____ property.

Choices:

- a. Text
- b. String
- c. Data
- d. mnemonic

Correct Answer:

a. Text

Explanation:

The Text property stores whatever the user types into a TextBox.

Question 2

A _____ is a storage location in memory that is represented by a name.

Choices:

- a. data type
- b. namespace
- c. variable
- d. cite

Correct Answer:

c. variable

Explanation:

A variable stores data in memory and is accessed using its name.

Question 3

In C#, you must _____ a variable before you can use it to store data.

Choices:

- a. associate
- b. instance
- c. declare
- d. reference

Correct Answer:

c. declare

Explanation:

Variables must be declared with a data type before use.

Question 4

A variable's _____ indicates the type of data that the variable will hold.

Choices:

- a. name
- b. data type
- c. scope
- d. value

Correct Answer:

b. data type

Explanation:

The data type defines what kind of value the variable can store.

Question 5

Fundamental types of data, such as strings, integers, and real numbers, are known as _____.

Choices:

- a. primitive data types
- b. fundamental variables
- c. logical digits
- d. literal data types

Correct Answer:

a. primitive data types

Explanation:

Primitive data types are the basic built-in data types in C#.

Question 6

A _____ identifies a variable in the program code.

Choices:

- a. binary number
- b. variable name
- c. unique global identifier
- d. hexadecimal value

Correct Answer:

b. variable name

Explanation:

The variable name is used to reference the variable in code.

Question 7

A common operation performed on strings is _____, or appending one string to the end of another string.

Choices:

- a. addition
- b. merging
- c. concatenation
- d. tying

Correct Answer:

c. concatenation

Explanation:

Concatenation joins strings together using operators like +.

Question 8

A _____ belongs to the method in which it is declared, and only statements inside that method can access it.

Choices:

- a. method variable
- b. primitive variable
- c. temporary variable
- d. local variable

Correct Answer:

d. local variable

Explanation:

Local variables are accessible only within the method where they are declared.

Question 9

Programmers use the term _____ to describe the part of a program in which a variable may be accessed.

Choices:

- a. range
- b. scope
- c. focus
- d. field

Correct Answer:

b. scope

Explanation:

Scope defines where a variable can be used in a program.

Question 10

A variable's _____ is the time period during which the variable exists in memory while the program is executing.

Choices:

- a. lifetime
- b. run time
- c. time to live
- d. half life

Correct Answer:

a. lifetime

Explanation:

Lifetime refers to how long a variable exists in memory.

Question 11

One way to make sure that a variable has been assigned a value is to _____ the variable when you declare it.

Choices:

- a. concatenate
- b. initialize
- c. delimit
- d. restrict

Correct Answer:

b. initialize

Explanation:

Initialization assigns an initial value to a variable at declaration.

Question 12

You can use a _____ to explicitly convert a value from one numeric data type to another.

Choices:

- a. transpose statement
- b. cast operator
- c. conversion operator
- d. literal conversion

Correct Answer:

b. cast operator

Explanation:

Casting forces a value to be treated as another data type.

Question 13

The process of dropping a number's fractional part is called _____.

Choices:

- a. shifting

- b. twos complement
- c. numeric rounding
- d. truncation

Correct Answer:

d. truncation

Explanation:

Truncation removes the decimal part without rounding.

Question 14

A programmer's tools for performing calculations are _____.

Choices:

- a. math operators
- b. numeric literals
- c. local variables
- d. parsed literals

Correct Answer:

a. math operators

Explanation:

Math operators such as +, -, *, and / perform calculations.

Question 15

A _____ performs a calculation and gives a value.

Choices:

- a. numeric literal
- b. math expression
- c. machine instruction
- d. programming statement

Correct Answer:

b. math expression

Explanation:

Math expressions combine values and operators to produce a result.

Question 16

C# offers a special set of operators designed for changing a variable's value without typing its name twice, called _____.

Choices:

- a. combined assignment operators
- b. advanced math operators
- c. variable modifiers
- d. assignment sequencers

Correct Answer:

a. combined assignment operators

Explanation:

Operators like += and -= update values efficiently.

Question 17

In computer science, the term _____ typically means to analyze a string of characters.

Choices:

- a. compile
- b. compute
- c. debug
- d. parse

Correct Answer:

d. parse

Explanation:

Parsing examines text to extract or interpret data.

Question 18

A(n) _____ is a piece of data that is passed into a method.

Choices:

- a. variable
- b. argument
- c. string
- d. literal

Correct Answer:

b. argument

Explanation:

Arguments provide input values to methods.

Question 19

A(n) _____ is an unexpected error that occurs while a program is running.

Choices:

- a. breakpoint
- b. bug
- c. syntax error
- d. exception

Correct Answer:

d. exception

Explanation:

Exceptions occur at runtime and may stop program execution if unhandled.

Question 20

The _____ indicates that you want a number formatted in a specific way when returned as a string.

Choices:

- a. formatting string
- b. insert method
- c. data type
- d. variable name

Correct Answer:

a. formatting string

Explanation:

Formatting strings like "C" or "N" control output format.

True / False**Statement 1**

You can clear the contents of a TextBox control the same way you clear a Label control.

Correct Answer: True

Explanation:

Both controls use the Text property, which can be set to an empty string.

Statement 2

In C#, you must declare a variable before you can use it.

Correct Answer: True

Explanation:

Variables must be declared with a data type before use.

Statement 3

You can declare multiple variables of different data types with one declaration.

Correct Answer: False

Explanation:

Only variables of the same data type can be declared in a single statement.

Statement 4

Appending the letter D or d to a numeric literal makes it a decimal literal.

Correct Answer: False

Explanation:

D or d creates a double literal; decimal literals use M or m.

Statement 5

The order of operations requires division to be performed before addition.

Correct Answer: True

Explanation:

Division has higher precedence than addition.

Statement 6

All variables have a ToString method.

Correct Answer: True

Explanation:

All data types inherit ToString from the base Object class.

Statement 7

Using the formatting string "C" formats a number as currency.

Correct Answer: True

Explanation:

"C" or "c" formats numeric values as currency.

Statement 8

When you declare a named constant, an initialization value is required.

Correct Answer: True

Explanation:

Constants must be assigned a value at declaration.

Statement 9

An error occurs if code attempts to change the value of a constant field.

Correct Answer: True

Explanation:

Constants cannot be modified after declaration.

Statement 10

Forms and controls have a Preferences property for changing background color.

Correct Answer: False

Explanation:

The correct property is BackColor, not Preferences.

Question 1

A _____ structure executes a set of statements only under certain circumstances.

Choices:

- a. circumstantial
- b. decision
- c. Boolean
- d. sequence

Correct Answer:

b. decision

Explanation:

A decision structure allows a program to choose whether to execute certain statements based on a condition.

Question 2

A _____ structure provides one alternative path of execution.

Choices:

- a. single-alternative decision
- b. one-path alternative
- c. single-execution decision
- d. binary

Correct Answer:

a. single-alternative decision

Explanation:

A single-alternative decision executes code only if a condition is true.

Question 3

A(n) _____ expression has a value of either true or false.

Choices:

- a. decision
- b. unconditional
- c. Boolean
- d. relational

Correct Answer:

c. Boolean

Explanation:

Boolean expressions evaluate to either true or false.

Question 4

The symbols >, <, and == are all _____ operators.

Choices:

- a. logical
- b. conditional
- c. ternary
- d. relational

Correct Answer:

d. relational

Explanation:

Relational operators compare values and produce Boolean results.

Question 5

A _____ structure tests a condition and then takes one path if the condition is true and another path if the condition is false.

Choices:

- a. multibranch statement
- b. single-alternative decision
- c. dual-alternative decision
- d. sequence

Correct Answer:

c. dual-alternative decision

Explanation:

A dual-alternative decision provides two possible execution paths.

Question 6

You use a(n) _____ statement to write a single-alternative decision structure.

Choices:

- a. test-jump

- b. if
- c. if-else
- d. if-call

Correct Answer:

b. if

Explanation:

The if statement executes code only when its condition is true.

Question 7

You use a(n) _____ statement to write a dual-alternative decision structure.

Choices:

- a. test-jump
- b. if
- c. if-else
- d. if-call

Correct Answer:

c. if-else

Explanation:

The if-else statement provides two alternative execution paths.

Question 8

A _____ decision structure is written inside another decision structure.

Choices:

- a. nested
- b. tiered

- c. dislodged
- d. hierarchical

Correct Answer:

a. nested

Explanation:

Nested decisions are placed inside other decision structures.

Question 9

&&, ||, and ! are _____ operators.

Choices:

- a. relational
- b. logical
- c. conditional
- d. ternary

Correct Answer:

b. logical

Explanation:

Logical operators combine or modify Boolean expressions.

Question 10

A compound Boolean expression created with the && operator is true only if _____.

Choices:

- a. both subexpressions are true
- b. either subexpression is true
- c. both subexpressions are false
- d. neither subexpression is evaluated

Correct Answer:

a. both subexpressions are true

Explanation:

The AND (&) operator requires both conditions to be true.

Question 11

A compound Boolean expression created with the || operator is true if _____.

Choices:

- a. both subexpressions are true
- b. either subexpression is true
- c. both subexpressions are false
- d. neither subexpression is evaluated

Correct Answer:

b. either subexpression is true

Explanation:

The OR (||) operator returns true if at least one condition is true.

Question 12

The _____ operator takes a Boolean expression and reverses its logical value.

Choices:

- a. &&
- b. ||
- c. !
- d. either

Correct Answer:

c. !

Explanation:

The NOT (!) operator negates a Boolean value.

Question 13

A _____ is a Boolean variable that signals when some condition exists in a program.

Choices:

- a. flag
- b. signal
- c. sentinel
- d. siren

Correct Answer:

a. flag

Explanation:

Flags are commonly used to indicate whether a condition has occurred.

Question 14

The _____ family of methods converts a string to a data type without throwing an exception.

Choices:

- a. TryConvert
- b. Parse
- c. TryParse
- d. SafeConvert

Correct Answer:

c. TryParse

Explanation:

TryParse returns true or false instead of throwing an exception.

Question 15

If several _____ controls exist in a GroupBox, only one may be selected at a time.

Choices:

- a. CheckBox
- b. RadioButton
- c. ListBox
- d. SelectionButton

Correct Answer:

b. RadioButton

Explanation:

RadioButtons within the same GroupBox are mutually exclusive.

Question 16

You use the _____ statement to create a multiple-alternative decision structure.

Choices:

- a. menu
- b. branch
- c. select
- d. switch

Correct Answer:

d. switch

Explanation:

The switch statement allows multiple execution paths.

Question 17

The _____ section of a switch statement executes when no case matches.

Choices:

- a. else
- b. default
- c. case
- d. otherwise

Correct Answer:

b. default

Explanation:

The default section handles unmatched cases.

Question 18

A ListBox's index numbering starts at _____.

Choices:

- a. 0
- b. 1
- c. -1
- d. any value

Correct Answer:

a. 0

Explanation:

ListBox items are zero-indexed.

Question 19

Which property indicates whether an item is selected in a ListBox?

Choices:

- a. Index
- b. SelectedItem
- c. SelectedIndex
- d. Items.SelectedIndex

Correct Answer:

c. SelectedIndex

Explanation:

SelectedIndex shows the index of the selected item.

Question 20

Which property holds the selected item in a ListBox?

Choices:

- a. Index
- b. SelectedItem
- c. SelectedIndex
- d. Items.SelectedIndex

Correct Answer:

b. SelectedItem

Explanation:

SelectedItem stores the actual selected object.

True / False**Statement 1**

You can write any program using only sequence structures.

Correct Answer: False

Explanation:

Decision and repetition structures are also required.

Statement 2

A single-alternative decision structure has two possible execution paths.

Correct Answer: False

Explanation:

It has only one path that executes when the condition is true.

Statement 3

The if-else statement is a dual-alternative decision structure.

Correct Answer: True

Explanation:

It provides exactly two execution paths.

Statement 4

A decision structure can be nested inside another decision structure.

Correct Answer: True

Explanation:

Nested decisions are allowed and commonly used.

Statement 5

The && operator is true only when both conditions are true.

Correct Answer: True

Explanation:

This is the definition of the logical AND operator.

Statement 6

TryParse methods throw an exception if conversion fails.

Correct Answer: False

Explanation:

TryParse returns false instead of throwing an exception.

Statement 7

Multiple CheckBox controls in the same GroupBox can be selected at the same time.

Correct Answer: True

Explanation:

CheckBox controls are independent of each other.

Statement 8

The test expression in a switch statement can be a double value.

Correct Answer: False

Explanation:

Switch expressions typically use int, char, string, or enum types.

Statement 9

If no item is selected in a ListBox, SelectedIndex is set to 0.

Correct Answer: False

Explanation:

If no item is selected, SelectedIndex is set to -1.

Statement 10

Items are added to a ListBox using its Text property.

Correct Answer: False

Explanation:

Items are added using the Items collection, not the Text property.

Question 1

ListBox controls have an _____ method that erases all the items in the Items property.

Choices:

- a. Items.Remove
- b. Items.Clear
- c. Items.Empty
- d. counter variable

Correct Answer:

b. Items.Clear

Explanation:

The Items.Clear method removes all items from a ListBox.

Question 2

A _____ is commonly used to control the number of times that a loop iterates.

Choices:

- a. test expression
- b. while clause
- c. controlled variable
- d. preemptive

Correct Answer:

c. controlled variable

Explanation:

A controlled (counter) variable tracks how many times a loop executes.

Question 3

A(n) _____ loop tests its condition before performing an iteration.

Choices:

- a. pretest
- b. infinite
- c. logical
- d. input file

Correct Answer:

a. pretest

Explanation:

A pretest loop checks its condition before executing its body.

Question 4

The term _____ is used to describe a file that data is written to.

Choices:

- a. input file
- b. output file
- c. saved file
- d. user file

Correct Answer:

b. output file

Explanation:

An output file is a file to which data is written.

Question 5

The term _____ file is used to describe a file that data is read from.

Choices:

- a. data
- b. write
- c. read
- d. input

Correct Answer:

d. input

Explanation:

An input file provides data to a program.

Question 6

A _____ file contains data that has been encoded as text, using a scheme such as Unicode.

Choices:

- a. text
- b. character
- c. Unicode
- d. system

Correct Answer:

a. text

Explanation:

Text files store data as readable characters.

Question 7

When you work with a _____ file, you access data from the beginning of the file to the end.

Choices:

- a. direct access
- b. random access
- c. sequential access
- d. binary access

Correct Answer:

c. sequential access

Explanation:

Sequential access reads data in order from start to finish.

Question 8

Files on a disk are identified by a _____.

Choices:

- a. unique identifier
- b. filename
- c. binary sequencer
- d. file extension

Correct Answer:

b. filename

Explanation:

A filename uniquely identifies a file on disk.

Question 9

A _____ object is associated with a specific file and allows a program to work with that file.

Choices:

- a. data
- b. directory
- c. stream
- d. file

Correct Answer:

c. stream

Explanation:

Streams provide a way to read from or write to files.

Question 10

When a program works with an input file, a special value that marks the location of the next item to be read is called the _____.

Choices:

- a. input locator

- b. accumulator
- c. read position
- d. sequential read value

Correct Answer:

c. read position

Explanation:

The read position tracks where the next read will occur.

Question 11

When the user selects a file using the Open dialog box, the file's path and name are stored in the control's _____ property.

Choices:

- a. Filename
- b. FilePath
- c. Pathname
- d. Text

Correct Answer:

d. Text

Explanation:

The Text property stores the selected file's path and name.

Question 12

The _____ control displays a standard Windows Save As dialog box.

Choices:

- a. SaveAsDialog
- b. FileDialog

- c. SaveFileDialog
- d. StandardDialog

Correct Answer:

c. SaveFileDialog

Explanation:

SaveFileDialog allows users to choose a file name and location for saving.

Question 13

Once you have created a Random object, you call its _____ method to generate a random integer.

Choices:

- a. Generate
- b. Rand
- c. NextInteger
- d. Next

Correct Answer:

d. Next

Explanation:

The Next method returns pseudo-random numbers.

Question 14

When generating random numbers, the _____ value is used in the calculation that produces the next number.

Choices:

- a. Start
- b. seed

- c. Next
- d. sequence

Correct Answer:

b. seed

Explanation:

The seed initializes the random number generator.

Question 15

When an application starts, the form is loaded into memory and the _____ event occurs.

Choices:

- a. Startup
- b. Begin
- c. Load
- d. Initialize

Correct Answer:

c. Load

Explanation:

The Load event fires when the form is first loaded.

True / False

Statement 1

If a ListBox is empty, the Items.Count property equals -1.

Correct Answer: False

Explanation:

Items.Count equals 0 when the ListBox is empty.

Statement 2

The while loop is a pretest loop.

Correct Answer: True

Explanation:

The while loop checks its condition before executing.

Statement 3

To increment a variable means to increase its value; to decrement means to decrease it.

Correct Answer: True

Explanation:

Increment increases by 1; decrement decreases by 1.

Statement 4

When a variable is declared in a for loop, its scope is limited to that loop.

Correct Answer: True

Explanation:

Variables declared in the for loop exist only within the loop.

Statement 5

The while loop always executes at least one time.

Correct Answer: False

Explanation:

The while loop may execute zero times if the condition is false.

Statement 6

The term read file describes a file that data is read from.

Correct Answer: False

Explanation:

The correct term is input file.

Statement 7

To append data to a file, you open it using the File.AppendText method.

Correct Answer: True

Explanation:

AppendText adds data to the end of a file.

Statement 8

As data is read from a file, the read position moves forward.

Correct Answer: True

Explanation:

The read position advances after each read operation.

Statement 9

Random numbers generated by the Random class are truly random.

Correct Answer: False

Explanation:

They are pseudo-random numbers.

Statement 10

The Load event occurs after the form is displayed.

Correct Answer: False

Explanation:

The Load event occurs before the form is displayed.

Chapter 5

Modularizing Your Code with Methods

Question 1

In general terms, a program that is broken into smaller units of code, such as methods, is known as a(n) _____.

Choices:

- a. object-oriented program
- b. modularized program
- c. procedural program
- d. method-driven program

Correct Answer:

b. modularized program

Explanation:

A modularized program is divided into smaller, manageable units (methods) to improve readability and maintenance.

Question 2

Writing code once and reusing it whenever needed is a benefit of using methods known as _____.

Choices:

- a. code reuse
- b. the single-use philosophy
- c. method recycling
- d. code reprocessing

Correct Answer:

a. code reuse

Explanation:

Methods allow the same code to be executed multiple times without rewriting it.

Question 3

When you call a(n) _____, it executes its statements and then terminates without returning a value.

Choices:

- a. intrinsic method
- b. empty method
- c. logical method
- d. void method

Correct Answer:

d. void method

Explanation:

A void method performs actions but does not return a value.

Question 4

The _____ appears at the beginning of a method definition and includes the method's name.

Choices:

- a. method title
- b. method description
- c. method header
- d. method declaration

Correct Answer:

c. method header

Explanation:

The method header specifies important information such as return type and method name.

Question 5

The _____ is the collection of statements enclosed in curly braces that execute when the method runs.

Choices:

- a. method body
- b. method designation
- c. method code
- d. method classification

Correct Answer:

a. method body

Explanation:

The method body contains the executable statements.

Question 6

The _____ is the memory address saved when a method is called and indicates where execution should return.

Choices:

- a. calling address
- b. method address
- c. return point
- d. come back position

Correct Answer:

c. return point

Explanation:

The return point tells the program where to continue after the method finishes.

Question 7

Programmers commonly use _____ to break down an algorithm into methods.

Choices:

- a. prototyping
- b. method modeling
- c. algorithm division
- d. top-down design

Correct Answer:

d. top-down design

Explanation:

Top-down design starts with a high-level overview and breaks it into smaller tasks.

Question 8

Pieces of data that are sent into a method are known as _____.

Choices:

- a. arguments
- b. references
- c. method variables
- d. data entries

Correct Answer:

a. arguments

Explanation:

Arguments provide values to a method when it is called.

Question 9

A(n) _____ is a special variable that receives an argument when a method is called.

Choices:

- a. reference variable
- b. argument variable
- c. parameter variable
- d. method variable

Correct Answer:

c. parameter variable

Explanation:

Parameters receive values from arguments passed to the method.

Question 10

A _____ specifies which parameter an argument should be passed into.

Choices:

- a. named argument
- b. special argument
- c. constant argument
- d. literal argument

Correct Answer:

a. named argument

Explanation:

Named arguments explicitly indicate which parameter receives a value.

Question 11

When a(n) _____ is provided for a parameter, the method can be called without supplying that argument.

Choices:

- a. local argument
- b. empty argument
- c. default argument
- d. expressional argument

Correct Answer:

c. default argument

Explanation:

Default arguments supply a value if none is provided.

Question 12

When an argument is _____, only a copy of its value is passed to the method.

Choices:

- a. a named constant
- b. passed by association
- c. passed by reference
- d. passed by value

Correct Answer:

d. passed by value

Explanation:

Pass-by-value prevents the method from modifying the original variable.

Question 13

To allow a method to change the value of a variable passed to it, the variable must be _____.

Choices:

- a. passed by reference
- b. a local variable
- c. passed by value
- d. a named constant

Correct Answer:

a. passed by reference

Explanation:

Pass-by-reference allows direct modification of the original variable.

Question 14

A _____ is a special type of parameter used for two-way communication between methods.

Choices:

- a. communication variable
- b. reference parameter
- c. method parameter
- d. global variable

Correct Answer:

b. reference parameter

Explanation:

Reference parameters allow methods to send and receive values.

Question 15

A(n) _____ can receive an uninitialized value but must be assigned a value before the method ends.

Choices:

- a. input parameter
- b. reference parameter
- c. output parameter
- d. default parameter

Correct Answer:

c. output parameter

Explanation:

Output parameters must be assigned inside the method before it finishes.

Question 16

A method's _____ specifies the type of value it returns.

Choices:

- a. return type
- b. value type
- c. method type
- d. data type

Correct Answer:

a. return type

Explanation:

The return type defines what kind of value a method sends back.

True / False

Statement 1

Dividing a large problem into smaller problems is called divide and conquer.

Correct Answer: True

Explanation:

This approach simplifies complex problems.

Statement 2

In a PascalCase name, the first letter is uppercase.

Correct Answer: True

Explanation:

PascalCase requires each word to start with a capital letter.

Statement 3

A method that belongs to a class must be defined inside that class.

Correct Answer: True

Explanation:

Class methods must be written inside the class definition.

Statement 4

The contents of variables and expressions cannot be passed as arguments.

Correct Answer: False

Explanation:

Variables and expressions can be passed as arguments.

Statement 5

You do not need to specify a data type for each parameter if they are the same type.

Correct Answer: False

Explanation:

In C#, each parameter must explicitly declare its data type.

Statement 6

An output parameter behaves the same as a pass-by-value parameter.

Correct Answer: False

Explanation:

Output parameters allow values to be returned from a method.

Statement 7

A value-returning method must contain a return statement.

Correct Answer: True

Explanation:

A return statement is required to send a value back to the caller.

Statement 8

A Boolean method returns either yes or no.

Correct Answer: False

Explanation:

A Boolean method returns true or false, not yes or no.

Chapter 6

Arrays and Lists

Question 1

The memory that is allocated for a _____ variable is the actual location that holds the value assigned to it.

Choices:

- a. general type
- b. value type
- c. framework type
- d. reference variable

Correct Answer:

b. value type

Explanation:

A value type stores the actual data directly in memory.

Question 2

A variable that is used to reference an object is commonly called a(n) _____.

Choices:

- a. resource variable
- b. object variable
- c. component variable
- d. union

Correct Answer:

b. object variable

Explanation:

An object variable holds a reference to an object in memory.

Question 3

When you want to work with an object, you use a variable that holds a _____ to link the variable to the object.

Choices:

- a. reference
- b. object linker
- c. data coupling
- d. = operator

Correct Answer:

a. reference

Explanation:

A reference connects a variable to the object's memory location.

Question 4

The _____ creates an object in memory and returns a reference to that object.

Choices:

- a. object allocator
- b. reference variable
- c. new operator
- d. array

Correct Answer:

c. new operator

Explanation:

The new operator allocates memory and returns a reference.

Question 5

A(n) _____ is an object that can hold a group of values of the same data type.

Choices:

- a. collection
- b. array
- c. container
- d. set

Correct Answer:

b. array

Explanation:

Arrays store multiple values of the same type.

Question 6

The _____ indicates the number of values that an array can hold.

Choices:

- a. allocation limit
- b. size declarator
- c. data type
- d. compiler

Correct Answer:

b. size declarator

Explanation:

The size declarator specifies the array's capacity.

Question 7

The storage locations in an array are called _____.

Choices:

- a. elements
- b. sectors
- c. pages
- d. blocks

Correct Answer:

a. elements

Explanation:

Each value in an array is stored in an element.

Question 8

Each element in an array is assigned a unique number called a(n) _____.

Choices:

- a. element identifier
- b. subscript
- c. index
- d. sequencer

Correct Answer:

c. index

Explanation:

Indexes identify the position of elements in an array.

Question 9

When you create an array, you can optionally initialize it using a(n) _____.

Choices:

- a. default value group
- b. initialization list
- c. defined set
- d. value list

Correct Answer:

b. initialization list

Explanation:

Initialization lists assign values at array creation.

Question 10

In C#, all arrays have a _____ that stores the number of elements in the array.

Choices:

- a. Limit property
- b. Size property
- c. Length property
- d. Maximum property

Correct Answer:

c. Length property

Explanation:

The Length property gives the total number of elements.

Question 11

A(n) _____ occurs when a loop iterates one time too many or too few.

Choices:

- a. general error
- b. logic error
- c. loop count error
- d. off-by-one error

Correct Answer:

d. off-by-one error

Explanation:

Off-by-one errors are common loop boundary mistakes.

Question 12

C# provides a special loop that simplifies array processing, known as the _____.

Choices:

- a. for loop
- b. foreach loop
- c. while loop
- d. do-while loop

Correct Answer:

b. foreach loop

Explanation:

foreach automatically iterates through array elements.

Question 13

The foreach loop uses a temporary read-only variable called the _____.

Choices:

- a. element variable

- b. loop variable
- c. index variable
- d. iteration variable

Correct Answer:

a. element variable

Explanation:

The element variable represents each value in the collection.

Question 14

_____ is a process that removes unreferenced objects from memory.

Choices:

- a. Systematic reallocation
- b. Memory cleanup
- c. Garbage collection
- d. Object maintenance

Correct Answer:

c. Garbage collection

Explanation:

Garbage collection automatically frees unused memory.

Question 15

Techniques used to locate an item in a collection are called _____.

Choices:

- a. seek functions
- b. request methods

- c. traversal procedures
- d. search algorithms

Correct Answer:

d. search algorithms

Explanation:

Search algorithms are designed to find specific values.

Question 16

The _____ searches an array sequentially from the first element to the last.

Choices:

- a. sequential search algorithm
- b. top-down method
- c. ascending search algorithm
- d. basic search function

Correct Answer:

a. sequential search algorithm

Explanation:

Sequential search checks elements one by one.

Question 17

A(n) _____ copies only the reference to an array, not its contents.

Choices:

- a. object copy
- b. reference copy
- c. double reference
- d. parallel copy

Correct Answer:

b. reference copy

Explanation:

Both variables point to the same array in memory.

Question 18

The _____ is more efficient than a sequential search.

Choices:

- a. linear search
- b. bubble sort
- c. binary search
- d. selection sort

Correct Answer:

c. binary search

Explanation:

Binary search divides the search space in half each step.

Question 19

A _____ is similar to a two-dimensional array, but rows can have different lengths.

Choices:

- a. one-dimensional array
- b. columnar array
- c. jagged array
- d. split row array

Correct Answer:

c. jagged array

Explanation:

Jagged arrays allow variable-length rows.

Question 20

The .NET Framework provides a class named _____ for storing and retrieving items.

Choices:

- a. Matrix
- b. Database
- c. Container
- d. List

Correct Answer:

d. List

Explanation:

The List class stores dynamic collections of items.

True / False**Statement 1**

Value type variables store actual data values.

Correct Answer: True

Explanation:

Value types directly store their data.

Statement 2

Reference variables can only reference objects.

Correct Answer: True

Explanation:

Reference variables point to objects in memory.

Statement 3

Individual variables are well suited for storing lists of data.

Correct Answer: False

Explanation:

Arrays and lists are better suited for storing collections.

Statement 4

Arrays are reference type objects.

Correct Answer: True

Explanation:

Arrays are stored as reference types in C#.

Statement 5

An array can store values of different data types.

Correct Answer: False

Explanation:

All elements must be of the same data type.

Statement 6

Numeric arrays are initialized with zeros by default.

Correct Answer: True

Explanation:

Default numeric values are zero.

Statement 7

The index of the last array element is one less than the Length property.

Correct Answer: True

Explanation:

Array indexing starts at 0.

Statement 8

The == operator compares array contents.

Correct Answer: False

Explanation:

== compares references, not contents.

Statement 9

A jagged array allows rows of different lengths.

Correct Answer: True

Explanation:

This is the defining feature of jagged arrays.

Statement 10

You must know the number of items before creating a List object.

Correct Answer: False

Explanation:

Lists grow dynamically as items are added.

Chapter 7

More about Processing Data

(Characters, Strings, Structures, Enumerations, and ImageLists)

Question 1

In C#, _____ are enclosed in single quotation marks.

Choices:

- a. enumerators
- b. tokens
- c. character literals
- d. strings

Correct Answer:

c. character literals

Explanation:

Character literals represent a single character and are written between single quotes, such as 'A'.

Question 2

The char data type provides the _____ for converting the case of a character.

Choices:

- a. Lowercase and Uppercase methods
- b. ToLower and ToUpper methods
- c. IsDigit and IsLetter methods
- d. IsWhiteSpace and IsPunctuation methods

Correct Answer:

b. ToLower and ToUpper methods

Explanation:

These methods convert characters to lowercase or uppercase.

Question 3

A string within another string is called a(n) _____.

Choices:

- a. inner string
- b. substring
- c. thread
- d. character

Correct Answer:

b. substring

Explanation:

A substring is a portion of a string.

Question 4

The _____ of a string object allow you to search for substrings.

Choices:

- a. TrimStart and TrimEnd methods
- b. IndexOf and LastIndexOf methods
- c. IsWhiteSpace and IsPunctuation methods
- d. Contains, StartsWith, and EndsWith methods

Correct Answer:

d. Contains, StartsWith, and EndsWith methods

Explanation:

These methods determine whether a string contains or begins/ends with specific text.

Question 5

When you want to know the position of a substring, you use _____.

Choices:

- a. TrimStart or TrimEnd methods
- b. IndexOf or LastIndexOf methods
- c. ToLower or ToUpper methods
- d. StartsWith or EndsWith methods

Correct Answer:

b. IndexOf or LastIndexOf methods

Explanation:

These methods return the position of a substring in a string.

Question 6

The _____ of a string object retrieves a specific set of characters.

Choices:

- a. Substring method
- b. IndexOf method
- c. TrimStart and TrimEnd methods
- d. StartsWith and EndsWith methods

Correct Answer:

a. Substring method

Explanation:

Substring extracts a portion of a string.

Question 7

_____ are spaces that appear at the beginning of a string.

Choices:

- a. Leading spaces
- b. Primary spaces
- c. Starting spaces
- d. Empty spaces

Correct Answer:

a. Leading spaces

Explanation:

Leading spaces are whitespace characters before the text.

Question 8

_____ are spaces that appear at the end of a string.

Choices:

- a. Blank spaces
- b. Secondary spaces
- c. Ending spaces
- d. Trailing spaces

Correct Answer:

d. Trailing spaces

Explanation:

Trailing spaces appear after the text.

Question 9

A series of words or data items separated by characters in a string are called _____.

Choices:

- a. substrings
- b. elements
- c. characters
- d. tokens

Correct Answer:

d. tokens

Explanation:

Tokens are individual data items extracted from a string.

Question 10

The character that separates tokens is known as a _____.

Choices:

- a. partition
- b. literal
- c. delimiter
- d. symbol

Correct Answer:

c. delimiter

Explanation:

Delimiters define how tokens are separated.

Question 11

The process of breaking a string into tokens is called _____.

Choices:

- a. extracting
- b. tokenizing

- c. delimiting
- d. parsing

Correct Answer:

b. tokenizing

Explanation:

Tokenizing splits a string into individual tokens.

Question 12

The _____ file format is commonly used to export spreadsheet data to text files.

Choices:

- a. SDV
- b. CSV
- c. XML
- d. PDF

Correct Answer:

b. CSV

Explanation:

CSV (Comma-Separated Values) is widely used for data exchange.

Question 13

A _____ is a user-defined data type that contains one or more fields.

Choices:

- a. structure
- b. collection
- c. volume
- d. list

Correct Answer:

a. structure

Explanation:

Structures group related data into a single type.

Question 14

Before using a structure, you must create a(n) _____ of it.

Choices:

- a. copy
- b. instance
- c. declaration
- d. reference

Correct Answer:

b. instance

Explanation:

An instance allocates memory for the structure.

Question 15

When items in two data structures are related by index, a _____ exists.

Choices:

- a. binary union
- b. parallel relationship
- c. unilateral bond
- d. virtual connection

Correct Answer:

b. parallel relationship

Explanation:

Parallel arrays use matching indexes to relate data.

Question 16

When you create a(n) _____, you define symbolic values for that data type.

Choices:

- a. abstract data type
- b. symbolic data type
- c. enumerated data type
- d. cryptic data type

Correct Answer:

c. enumerated data type

Explanation:

Enums define named integer constants.

Question 17

_____ are constants that represent integer values in an enum.

Choices:

- a. Literals
- b. Enumerators
- c. Constants
- d. Tokens

Correct Answer:

b. Enumerators

Explanation:

Enumerators are named constants in an enumeration.

Question 18

The _____ is a container that stores multiple images.

Choices:

- a. ImageList control
- b. GroupBox control
- c. PictureBox control
- d. ComboBox control

Correct Answer:

a. ImageList control

Explanation:

ImageList stores images for use by other controls.

True / False**Statement 1**

You cannot store a string in a char variable.

Correct Answer: True

Explanation:

The char data type stores only a single character.

Statement 2

C# allows access to individual characters in a string using index notation.

Correct Answer: True

Explanation:

Strings can be accessed using brackets, such as text[0].

Statement 3

The Split method always divides a string into exactly two substrings.

Correct Answer: False

Explanation:

Split can produce multiple substrings.

Statement 4

All fields in a structure must be of the same data type.

Correct Answer: False

Explanation:

Structures can contain fields of different types.

Statement 5

You must create an instance of a structure before using it.

Correct Answer: True

Explanation:

An instance allocates memory for the structure.

Statement 6

Structures can only be passed to methods by reference.

Correct Answer: False

Explanation:

Structures can be passed by value or by reference.

Statement 7

You cannot directly compare structure objects.

Correct Answer: True

Explanation:

Structures do not support direct comparison unless overloaded.

Statement 8

Enumerators must be enclosed in quotation marks.

Correct Answer: False

Explanation:

Enumerators are identifiers, not strings.

Statement 9

An enum declaration can appear only inside a namespace.

Correct Answer: False

Explanation:

Enums can also be declared inside classes.

Statement 10

Enumerator values do not have to be unique.

Correct Answer: True

Explanation:

Multiple enumerators can share the same integer value.

Statement 11

Enumerators can be compared using relational operators.

Correct Answer: True

Explanation:

Enums are backed by integers and support comparisons.

Statement 12

ImageList controls are designed to store small images such as icons.

Correct Answer: True

Explanation:

ImageList is optimized for small images.

By: Maria